

Multiple-Model Estimators for Tracking Sharply Maneuvering Ground Targets

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This paper presents algorithms that can be used to track the state of kinematic point targets, moving in two dimensions, that are capable of making very sharp heading maneuvers over short periods of time. Ground vehicles with continuous wheel tracks moving in an open field may perform such maneuvers. These targets are capable of 60°/s turn rates, whereas measurements are received at only 1 Hz. We describe how an interacting multiple model estimator (IMM) should be designed to accommodate such maneuvers, and we introduce the nonzero mean, white noise turn-rate model to quickly detect sharp turns by modeling a multimode process noise probability density function. Results show that including such modes in an IMM achieves significant improvement in accuracy and consistency, as compared to conventional coordinated-turn IMM filters.

Manuscript received November 27, 2016; revised August 30, 2017; released for publication November 2, 2017. Date of publication January 12, 2018; date of current version June 7, 2018.

DOI: No. 10.1109/TAES.2018.2793019

Refereeing of this contribution was handled by F. Govaers.

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I. BACKGROUND ON THE SHARPLY MANEUVERING TARGET PROBLEM

Tracking of remote targets is a challenging task because measurement systems are inaccurate, they are unaware of the input forces that may be applied to targets, and the targets' dynamic properties cannot be modeled exactly. Some real-world targets, such as land vehicles with continuous wheel tracks, are capable of making very sharp, evasive maneuvers, effectively maintaining constant speed while making large changes in heading. Although a very different type of algorithm for zigzagging targets was developed in [1], little prior work has dealt with this specific type of maneuvering target. Targets with moderate and relatively smooth turn capability have been extensively studied.

The coordinated-turn (CT) dynamic model has been widely used for tracking air targets [2]. In this model, the turn rate $\omega(t)$ is a state and, in continuous time, evolves as a Wiener process (it is sometimes referred to as the Wiener-process turn rate model [3]). The turn rate results in lateral accelerations (normal to velocity) that dominate the maneuvering capability of the target. However, there are some issues with this model when it is considered for tracking sharply turning targets. First, the most popular discretization of the CT model approximates the Wiener process as a random walk, and this approximation degrades with decreased measurement rate and increased target maneuverability. An alternate discretization is proposed in [4], but it is still approximate. A white noise turn rate model (WNTR) that circumvents this issue is proposed here for dealing with sharply maneuvering targets.

Kinematic targets that experience accelerations normal to their velocity can be described by nonlinear recursive equations. Similarly, when measurement systems report polar coordinates, they are described by nonlinear functions of the target state. In the current application, the dynamic equation contains the nonlinear effect of a turn rate $\omega(t)$ either as a state to be estimated or as a stochastic input. The magnitude of the angle that the target's velocity vector can turn over a sampling interval is quite high (60°, for example) and this results in relatively high nonlinearity. The linear minimum mean-square error (MSE) filter for state estimation, the Kalman filter, is not optimal in such situations. The most common suboptimal solution, the extended Kalman filter (EKF), uses first-order Taylor series approximations of the dynamic and/or measurement equations [3]. Second-order correction terms were first specified in [5], and Bar-Shalom *et al.* [6] have generalized the algorithm to handle nonlinear stochastic inputs and have provided computational shortcuts as well. Due to its great performance, mathematical tractability, and since it requires no intrinsic design parameters (other than the parameters of the dynamic and measurement models), the second-order EKF (SOEKF) is used here. There are, however, many other choices for nonlinear filters. The unscented filter [7] is a computationally efficient method that utilizes numerical differentiation to estimate the first two moments of the state. The particle filter [8] runs Monte Carlo simulations at each time step to estimate and

propagate an approximate, non-Gaussian posterior state probability density function (PDF) using Bayes' Theorem. The Cubature filter [9] uses numerical integration and Bayes' Theorem to find the posterior state PDF, and only the first two moments are carried to the next time step. To handle the nonlinear measurement equation, we can opt for any of the previously mentioned methods, or we can convert the measurements from polar to Cartesian coordinates. A method of unbiased conversions, which includes second-order corrections, is given in [10] and is used here.

It is stated in [7] that a motivation of the unscented filter is the need to calculate Jacobians and Hessians in the SOEKF. However, once they are calculated, they are very convenient and accurate, and the SOEKF does not require any design parameters related to its first- and second-order Taylor series approximations. We show a simple derivation of the Jacobians and Hessians in the appendix.

Alone, the nonlinear methods described above do not implicitly handle the nonpredictable nature of random inputs that are dealt with in the realm of maneuvering target tracking. For example, if a kinematic target moves in a straight line for a long time, then suddenly executes a sharp turn, it is not ideal to model the turn rate process noise as an integral of Gaussian white noise (a Wiener process) because this integration implies memory and low-pass response in a target's turn behavior, and the sharp-turn maneuvers do not exhibit such Wiener process behavior. One of the first algorithms for dealing with maneuvering targets used statistical decision theory to augment the state of a constant-velocity model with an acceleration variable when maneuvers are detected [11]. Input estimation, which uses a history of measurement residuals to detect and estimate the magnitude of a step function input, was developed in [12].

When a target exhibits changes in the nature of its dynamics, then the system parameters can be modeled as Markov jump processes, and a unified framework of estimating both state and parameters can be used, as in [13] and [14]. Multiple-model (MM) algorithms, where the target's current system equation is one in a set of known "modes," are a modern choice for tracking maneuvering targets. The optimal MM estimator, along with the first suboptimal approximation (GPB1, the generalized pseudo-Bayesian MM estimator of first order), was presented in [15]. The GPB2 algorithm was first given in [16]. The interacting multiple model estimator (IMM), which typically performs as well as GPB2 but with much less computation, was first introduced in [17]. Since its introduction, it has generally been accepted as the best overall algorithm for maneuvering target tracking (and other MM system applications) when considering both accuracy and computational load [3], [18], [19], and it is used in this paper. Since then, the Viterbi algorithm has also been adapted to the area of MM switching systems [20], and a B-Best mode history algorithm [13] has also been developed. However, while theoretically sound, the latter two algorithms are hard-decision based and were shown [21] to have weak performance when the true target model is not among the models in the set designed for the algorithms. Variable-structure MM estimation, introduced

in [22], has been heavily researched in the past decade, but its inclusion into the present problem is unnecessary due to the small number of modes in the model sets that are presented here. A very different approach was taken in [23], where a very unpredictable target is tracked using a non-Bayesian segmenting algorithm.

A thorough survey of maneuvering target tracking can be found in [2], [24]–[26], and [19]. In that survey, as well as in [18] and [3], it is often the case that the best algorithms (considering both accuracy and computational load) for tracking point targets with CT capability in two dimension (2D) are IMM having a low-process-noise white noise acceleration (WNA) mode and one or more nearly CT modes (also known as Wiener process turn rate modes) in the model sets. In 2D, such an algorithm is known as IMM-CT. In [27]–[29], significant improvements were made to IMM that include mode models with state vectors that do not contain the same state variables. The work of Tugnait [29] encompasses the other two, and we will use it to correctly design prior distributions to be used in mixing the modes that contain the turn rate state $\omega(t)$ with those that do not.

Some of the filters presented here rely on nonzero-mean process noise models to represent straight-, left-, and right-turning target modes. This corresponds to each mode having an approximately known input. This method has been explored only in part in the literature. It was successfully applied in [21] by using multiple left- and right-turn modes, but there was no indication that this was a better method than using an IMM-CT with multiple maneuvering modes. Also, Linder and Schell [21] treated the turn rate in each mode as precisely known with no variance—this probabilistically incorrect in the present application. This paper generalizes the concept of nonzero-mean process noise to handle the quick, sharp turns that some ground vehicles can perform, without assuming that the turn rate is known exactly in any mode.

This paper is organized as follows. The problem formulation is given in Section II. Modifications to the IMM-CT and the new *white noise turn rate IMM* (IMM-WNTR) are presented in Section III to effectively track sharply maneuvering targets. Section IV formulates filter design parameter selection, and Section V discusses Monte Carlo simulation results. The appendix provides a short derivation of the Jacobians and Hessians of the CT and WNTR models.

II. PROBLEM FORMULATION

A. State and Measurement Equations

The target is modeled as a point in a 2-D Cartesian space with state vector

$$\begin{aligned} \mathbf{x}(t) &= [\mathbf{s}'(t) \quad \dot{\mathbf{s}}'(t) \quad \omega(t)]' \\ &= [x(t) \quad y(t) \quad \dot{x}(t) \quad \dot{y}(t) \quad \omega(t)]' \end{aligned} \quad (1)$$

where $\mathbf{s}(t) = [x(t) \quad y(t)]'$ is the Cartesian position, $\dot{\mathbf{s}}(t) = [\dot{x}(t) \quad \dot{y}(t)]'$ is the Cartesian velocity, and $\omega(t)$ is the turn rate of the velocity vector. In this paper, ω is a state variable only in the CT model. Our goal is to find the most accurate

estimates of the state $\hat{\mathbf{x}}(k|k)$ from noisy position measurements. We are also interested in finding an estimate of the MSE of the state estimate $\mathbf{P}(k|k)$. We are seeking higher accuracy than that can be obtained from the raw measurement system and consistency in the estimated MSE.

During a CT, the velocity vector rotates according to the turn rate ω . To model sharp maneuvers, such as those that can be undertaken by continuously tracked land vehicles deliberately attempting to evade conventional tracking systems and/or incoming weapons, we shall model and simulate high CT rates for short time periods. Positive turn rate implies counter-clockwise (left) rotation of the velocity vector.

The dynamic state evolution is described by the discrete-time system

$$\mathbf{x}_{k+1} = \mathbf{f}[\mathbf{x}_k, \mathbf{v}_k] \quad (2)$$

where $\mathbf{f}$ is the nonlinear state transition function between samples, and $\mathbf{v}(k)$ is a white process noise vector that represents random inputs or the model's inaccuracies. The process noise vector $\mathbf{v}_k$ has mean $\bar{\mathbf{v}}$ and covariance matrix $\mathbf{Q}$.

The measurement equation in polar coordinates is

$$\mathbf{z}_p(k) = [r(k) \ \theta(k)]' = \mathbf{h}(\mathbf{x}_s(k), \mathbf{x}(k)) + \mathbf{w}_p(k) \quad (3)$$

where $\mathbf{z}_p(k)$ is the vector that represents the noisy polar measurements of range $r(k)$ and angle $\theta(k)$, and $\mathbf{x}_s(k)$ is the state vector of the measurement sensor. The 2-D, additive, zero-mean, white, Gaussian vector $\mathbf{w}_p(k)$ describes the noise in the measurement, and

$$\mathbf{R}_p(k) = E[\mathbf{w}_p(k)\mathbf{w}_p'(k)] = \begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\theta^2 \end{bmatrix} \quad (4)$$

is its covariance matrix.

The target motion is expected to consist mostly of straight-line motion, fast left/right turning capability, slow turning capability, fast deceleration (braking) capability, and relatively slow acceleration capability. Although established methods can be used, such as IMM-CT and its variants, these methods were mostly designed for aerospace applications, where the maximum turns over a single sampling interval are only a few degrees, and where the turn rate itself cannot suddenly change because aircraft must bank to turn. However, it is critical that a filter can detect maneuvers and correct the state estimate as quickly as possible in the present application. It is also important to achieve very low tracking error during straight-line motion, where the dynamics are linear. All this should be done while maintaining the tracking accuracy better than the raw measurements.

III. MODELS FOR SHARPLY MANEUVERING TARGETS

Our algorithms are based on MM methods. For each filter, n_r will be the number of modes in the model set, and the first mode for all filters will use the nonmaneuvering WNA model. This linear model is typical for targets that perform only occasional accelerations and can be reviewed in [3].

A. CT Model

The CT model has been highly successful for tracking targets that are capable of true CTs, such as aircraft. Omitting the time parameter, the continuous-time model is

$$\begin{bmatrix} \dot{\mathbf{s}} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 2} & \omega \mathbf{U} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{1 \times 2} & \mathbf{0}_{1 \times 2} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{s} \\ \omega \end{bmatrix} + \begin{bmatrix} \mathbf{0}_{2 \times 3} \\ \mathbf{I}_{3 \times 3} \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_\omega \end{bmatrix} \quad (5)$$

where $\mathbf{I}_{m \times n}$ is the $m \times n$ identity matrix, $\mathbf{0}_{m \times n}$ is the $m \times n$ matrix of zeros, and

$$\mathbf{U} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (6)$$

is a constant skew-symmetric matrix. The random sequences $v_x(t)$, $v_y(t)$, and $v_\omega(t)$ are the components of the process noise $\mathbf{v}(t)$, with $v_\omega(t)$ the white noise turn acceleration that enters the system to yield a random walk (i.e., an approximate Wiener process) turn rate.

In discrete time, (5) is typically approximated as

$$\begin{bmatrix} \mathbf{s}_{k+1} \\ \dot{\mathbf{s}}_{k+1} \\ \omega_{k+1} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_{2 \times 2} & \frac{1}{\omega_k} \mathbf{U}(\mathbf{I} - \mathbf{V}_k) & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 2} & \mathbf{V}_k & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{1 \times 2} & \mathbf{0}_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{s}_k \\ \dot{\mathbf{s}}_k \\ \omega_k \end{bmatrix} + \begin{bmatrix} \frac{T^2}{2} \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ T \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{1 \times 2} & T \end{bmatrix} \begin{bmatrix} v_{x,k} \\ v_{y,k} \\ v_{\omega,k} \end{bmatrix} \quad (7)$$

where T is the measurement interval, $e^{\mathbf{A}}$ is the matrix exponential of $\mathbf{A}$, and

$$\mathbf{V}_k = e^{\omega_k T \mathbf{U}} = \begin{bmatrix} \cos \omega_k T & -\sin \omega_k T \\ \sin \omega_k T & \cos \omega_k T \end{bmatrix} \quad (8)$$

is a rotation matrix. To use (7), the turn rate ω_k is evaluated at the last estimate of the turn rate $\hat{\omega}_{k|k}$. It is evident that the evolution of ω_k in (7) is a random walk approximation of the Wiener process in (5). The turn rate is assumed constant throughout the state prediction interval T , so effects of the turn acceleration v_k are not propagated to the position and velocity vectors within the sampling interval. This issue is discussed in [4], and an alternate discrete approximation is proposed. The discrete-time CT model given in (7) can also be improved by running a prediction-only filter for m time intervals of length $\frac{T}{m}$ in between measurements for more accurate integration, at the expense of more computation [3].

Even with effective nonlinear filtering methods, the continuous-time CT model itself is still not a match to reality. This is because the sharp turns are of short duration and, when they do happen, the turn rate values are nearly independent of previous ones. We need to detect these maneuvers very quickly, but we do not want to sacrifice too much accuracy when there are no turns. Just increasing the variance of the turn acceleration σ_ω^2 needlessly increases the filter bandwidth during nonmaneuvering, straight-line motion.

B. Nonzero Mean, WNTR

If we want to detect occasional sharp maneuvers among nearly constant velocity motion, then it helps to model the turn rate process noise as a Gaussian mixture density (possibly multimodal) consisting of three Gaussian terms with different means: no turn (zero mean), left turn (positive mean), and right turn (negative mean). With a recursive estimation algorithm that requires that the process noise is approximated as a white Gaussian process, this is not possible. Within a MM estimator, however, this is very straightforward. In fact, we can approximate any non-Gaussian PDF using a Gaussian mixture [30]. MM estimators, such as the IMM, also allow us to describe the time correlation between modes in the form of a transition probability matrix (TPM) [3]. We can match the TPM to the expected behavior of the target by using the mean sojourn time of mode i , $E[\tau_i]$, with,

$$p_{ii} = 1 - \frac{T}{E[\tau_i]} \quad (9)$$

being the probability of remaining in mode i . If $E[\tau_i] = T$, then $p_{ii} = 0$. That means that the sharply maneuvering mode is expected to last one sample and no longer.

To approximate multimodal PDFs for the turn rate process noise, as we need for the occasional sharp maneuver, we can utilize nonzero mean process noise modes in a MM estimator. We call a model of these modes the nonzero mean WNTR model. The state is

$$\mathbf{x}_k = [\mathbf{s}_k \quad \dot{\mathbf{s}}_k]' \quad (10)$$

and the motion is described by

$$\begin{bmatrix} \mathbf{s}_{k+1} \\ \dot{\mathbf{s}}_{k+1} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_{2 \times 2} & \frac{1}{\omega_k} \mathbf{U}(\mathbf{I} - \mathbf{V}_k) \\ \mathbf{0}_{2 \times 2} & \mathbf{V}_k \end{bmatrix} \begin{bmatrix} \mathbf{s}_k \\ \dot{\mathbf{s}}_k \end{bmatrix} + \begin{bmatrix} \frac{T^2}{2} \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ T \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 1} \end{bmatrix} \begin{bmatrix} v_{x,k} \\ v_{y,k} \\ v_{\omega,k} \end{bmatrix}. \quad (11)$$

In (11), $v_{\omega,k}$ is the white noise turn rate. The second-order EKF will be used to propagate the state prediction $\hat{\mathbf{x}}_{k+1|k}$ and the state covariance prediction $\mathbf{P}_{k+1|k}$, with $\omega_k = \bar{v}_\omega$, the mean of the white noise turn rate process noise. We propose that when tracking sharply maneuvering targets, it is advantageous to have at least two WNTR modes to correspond to right and left turns.

An advantage of this nonzero mean process noise in a MM filter is that the measurement prediction $\hat{\mathbf{z}}_{k+1|k}^{(i)}$ of each mode i is unique, unlike the IMM-CT, where if $\hat{\omega}_{k|k}$ is zero, then the mode-conditioned measurement prediction of the maneuvering CT mode $\hat{\mathbf{z}}_{k+1|k}^{(2)}$ is almost the same as that of the nonmaneuvering mode $\hat{\mathbf{z}}_{k+1|k}^{(1)}$, resulting in less ability to discriminate between modes. Utilizing WNTR modes guarantees that the mode-conditioned state predictions are unique at all times since the mean of the turn rate process noise of each mode is unique, which results in unique measurement predictions. Higher-magnitude measurement residuals (innovations) will result in smaller values of the

likelihood functions for modes that are not in effect, thereby increasing the tracking accuracy during straight-line motion and allowing for fast detection/correction of turning maneuvers.

Even though the WNTR modes are designed for very sharp maneuvers, this filter can also perform well during CTs, thanks to the mode mixing. For example, during a CT to the left, the estimator mixes in some of the left turn mode's output, resulting in a balance between sharp turns and CV. While it will be shown that this does provide very good position and velocity estimates, this method does not estimate the turn rate, which means that it is not suited for situations where long-term target predictions are needed. To remedy this, we can also include a CT mode with low process noise in the mode set. This should provide the benefit of fast maneuver detection, as well as having a running estimate of ω .

C. Mode Mixing With Prior Distributions

If we introduce a CT mode in the IMM, then we must handle the mixing of modes that have different state vector components. The CT mode includes the ω state component, while the WNA and WNTR models do not. According to the work in [29], we should use a prior distribution of the ω state variable when transitioning from WNA/WNTR to CT. We must select a prior mean $\tilde{\omega}_k^{(i)}$ and variance $\tilde{\sigma}_{\omega,k}^{2(i)}$ of the turn rate for each non-CT mode i by asking "If the previous mode was WNA/WNTR, then what is the expected turn rate when starting the CT mode, and how much variance goes with that expected turn rate?"

To answer this question, we first note that the WNA model has a definite zero turn rate at all times.

$$\begin{aligned} \tilde{\omega}_k^{(i_{\text{WNA}})} &= 0 \\ \tilde{\sigma}_{\omega,k}^{2(i_{\text{WNA}})} &= 0. \end{aligned} \quad (12)$$

When entering CT from WNTR, we have

$$\begin{aligned} \tilde{\omega}_k^{(i_{\text{WNTR}})} &= \bar{v}_\omega^{(i_{\text{WNTR}})} \\ \tilde{\sigma}_{\omega,k}^{2(i_{\text{WNTR}})} &= \sigma_\omega^{2(i_{\text{WNTR}})} \end{aligned} \quad (13)$$

meaning that the expected turn rate when exiting the WNTR mode is the mean of the white noise turn rate, and the variance of this prior is equal to the variance of the white noise turn rate. $\bar{v}_\omega^{(i_{\text{WNTR}})}$ and $\sigma_\omega^{2(i_{\text{WNTR}})}$ are design parameters that are selected for each WNTR mode.

IV. FILTER DESIGN

A. IMM-CT Design

The first filter to test is a simple IMM-CT with mode 1, a WNA mode, and mode 2, a CT mode, matched to the possibility that the target can sharply turn its velocity vector within a single measurement interval. In mode 1, we set $\sigma_x = \sigma_y$ to a low value of 0.1 m/s^2 to cover small deviations from CV (and we use these linear acceleration standard deviation parameters in all subsequent modes designed in this paper). In mode 2, the CT model mode, we keep $\sigma_x = \sigma_y = 0.1 \text{ m/s}^2$ and set $\sigma_\omega^{(2)} = 60^\circ/\text{s}^2$ to handle

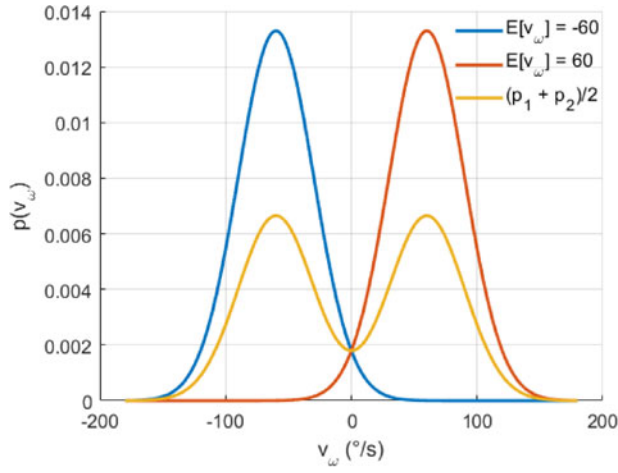


Fig. 1. Visualization of the white noise turn rate maneuvering process noise as a Gaussian sum mixture.

sharp-turn maneuvers. This parameter, the turn acceleration process noise standard deviation, is integrated as a constant over a measurement interval T to result in the target's most probable sharp-turn rate $\sigma_\omega T$, which is $60^\circ/\text{s}$ in this application. Initially, we set mean sojourn times as $E[\tau_1] = E[\tau_2] = 10$ s, which results in a typical IMM-CT design. This filter is labeled IMM-CT-S. Then, we form a second IMM to better match to the fast maneuvers by setting $E[\tau_2] = 1$ s, which results in $p_{22} = 0$, an unconventional IMM design. This filter is labeled IMM-CT-F.

B. White Noise Turn Rate IMM Design

The IMM-WNTR filter has a WNA mode (mode 1), a right-turn WNTR mode (mode 2), and a left-turn WNTR mode (mode 3). The model parameters are: $\bar{v}_\omega^{(2)} = -60^\circ/\text{s}$, $\bar{v}_\omega^{(3)} = 60^\circ/\text{s}$, and $\sigma_\omega^{(2)} = \sigma_\omega^{(3)} = 30^\circ/\text{s}$. We set $E[\tau_1] = 10$ s, $E[\tau_2] = E[\tau_3] = 1$ s. This gives $p_{11} = 0.9$ and $p_{22} = p_{33} = 0$. We split the remaining probabilities by setting the TPM to

$$\Pi_{\text{WNTR}} = \begin{bmatrix} p_{11} & \frac{1-p_{11}}{2} & \frac{1-p_{11}}{2} \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}. \quad (14)$$

Fig. 1 shows approximately what the maneuvering mode turn rate process noise looks like as a Gaussian sum mixture. Notice that by using the nonzero mean turn rate process noise, we can approximate a multimodal process noise PDF.

C. IMM-WNTR-CT Design

To design a filter that will also provide an estimate of the turn rate, we add a CT mode (fourth mode) to the IMM-WNTR to create the IMM-WNTR-CT. This filter has the same parameters as in Section IV-A, except $\sigma_\omega^{(4)} = 1^\circ/\text{s}^2$

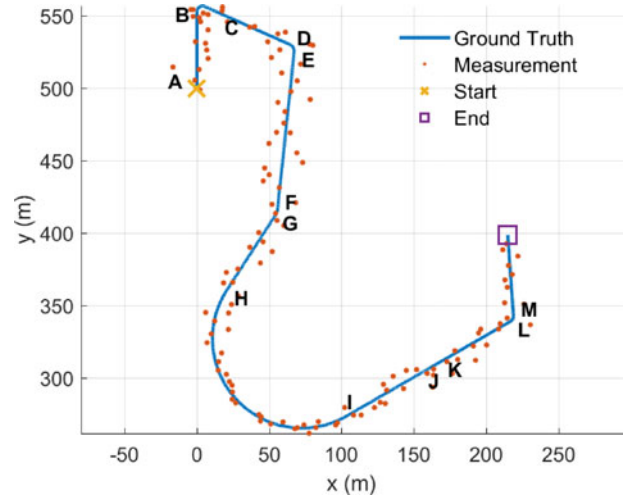


Fig. 2. Simulated trajectory and example measurements. Segment labels are from Table I.

and $E[\tau_4] = 10$ s. The TPM is

$$\Pi_{\text{CT-WNTR}} = \begin{bmatrix} p_{11} & \frac{1-p_{11}}{4} & \frac{1-p_{11}}{4} & \frac{1-p_{11}}{2} \\ 0.5 & 0 & 0 & 0.5 \\ 0.5 & 0 & 0 & 0.5 \\ 1-p_{44} & 0 & 0 & p_{44} \end{bmatrix}. \quad (15)$$

D. IMM-CT-LH Design

To ensure that adding two WNTR modes results in better performance than adding a single CT mode with high turn acceleration process noise, we will compare IMM-WNTR-CT to a three-mode IMM-CT-LH (referring to “Low” & “High” turn process noises, respectively). It contains a WNA mode (mode 1), a low process noise CT mode (mode 2), and a high process noise CT mode (mode 3). We set $\sigma_\omega^{(2)} = 1^\circ/\text{s}^2$, $\sigma_\omega^{(3)} = 60^\circ/\text{s}^2$, $E[\tau_2] = 10$ s, and $E[\tau_3] = 1$ s. The TPM is

$$\Pi_{\text{CT-LH}} = \begin{bmatrix} p_{11} & \frac{1-p_{11}}{2} & \frac{1-p_{11}}{2} \\ \frac{1-p_{22}}{2} & p_{22} & \frac{1-p_{22}}{2} \\ 0.5 & 0.5 & 0 \end{bmatrix}. \quad (16)$$

V. SIMULATIONS AND RESULTS

A. Simulated Trajectory and Measurements

For numerical simulations, we use a target trajectory shown in Fig. 2, along with an example of the noisy measurements. The sampling interval is $T = 1$ s. The target starts with $x_0 = 0$ m, $y_0 = 500$ m, $\dot{x}_0 = 0$ m/s, and $\dot{y}_0 = 5$ m/s. The simulated trajectory is designed to assess the algorithms' behavior for targets that perform very sharp turns, rapid decelerations, moderately sharp turns without deceleration, smooth CTs, and longitudinal acceleration. Randomly selected, fractional time intervals are used so that maneuvers do not necessarily start at the sampling instants. A description of the trajectory is given in Table I.

TABLE I
Trajectory Description With Maneuver Labels

Label	Start Time (s)	Duration (s)	Turn Rate ($^{\circ}/s$)	Long. Acc. (m/s^2)
A	0	10.5	0	0
B	10.5	1.9	-60 (R)	-1 (br)
–	12.4	2.3	0	1 (acc)
C	14.69	10.2	0	0
D	24.8	1.2	-60 (R)	0
E	26.08	20.2	0	0
F	46.28	0.9	-30 (R)	0
G	47.18	11.4	0	0
H	58.58	30.6	5 (L)	0
I	89.18	10.5	0	0
J	99.68	1.0	0	-5 (br)
–	100.68	5.2	0	1 (acc)
K	105.88	10	0	0
L	115.88	0.8	80 (L)	0
M	116.68	10	0	0

R = right turn, L = left turn, br = longitudinal braking maneuver, acc = longitudinal acceleration maneuver. Two maneuver labels are omitted to reduce visual clutter in the simulation results.

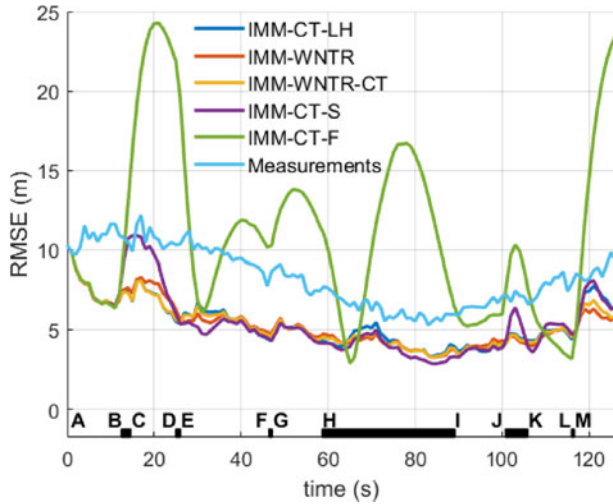


Fig. 3. Position estimate RMSE for 5 filters and raw measurements.

The total simulation time is 126.68 s. The measurements are made in polar coordinates r_k and θ_k with a sensor located at the origin. The measurement standard deviations are constant in polar coordinates and are given as $\sigma_r = 1.5$ m, $\sigma_{\theta} = 1.15^{\circ}$. The unbiased conversion of [10] is used to feed Cartesian measurements to all filters, resulting in all subsequent filter operations performed in Cartesian coordinates with position-dependent measurement variance; 1000 Monte Carlo runs are used for all performance evaluations.

B. RMSE and NEES Results

The position root-mean-square error (RMSE) across the five estimators and the raw measurements from Monte Carlo simulations is shown in Fig. 3. The velocity RMSE is shown in Fig. 4, and the normalized estimation error squared (NEES [3]) is shown in Fig. 5 for consistency evaluation. The NEES is calculated using only the position

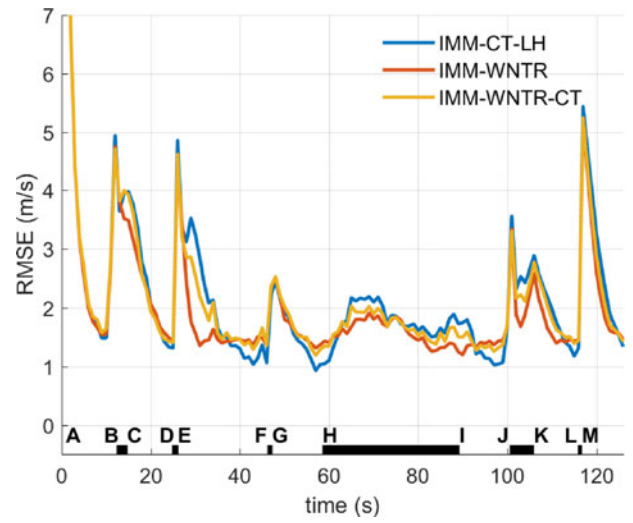


Fig. 4. Velocity estimate RMSE for five filters.

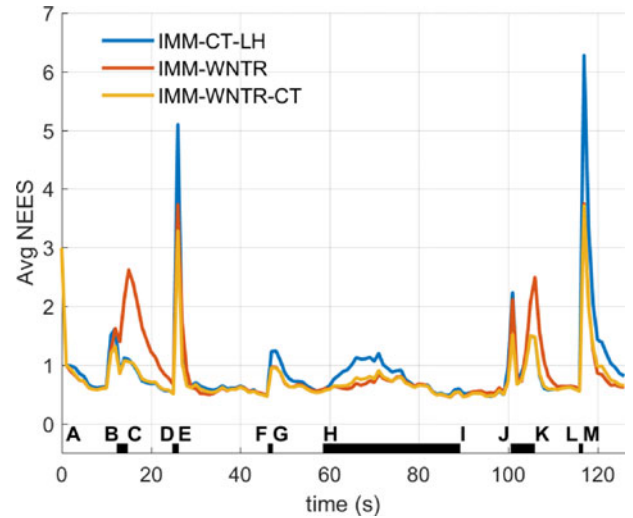


Fig. 5. NEES for five filters. Vertical axis is limited to 7, cutting off the IMM-CT-F and IMM-CT-S curves for better visibility of remaining filters.

and velocity state components, and it is normalized by 4 to achieve a nominal value of 1 for ideal consistency. Maneuvering intervals are drawn with black rectangles along the time axis, referenced by the trajectory segment labels in Table I. It can be seen from Fig. 3 that all filters, except IMM-CT-S and IMM-CT-F, achieve position RMSE that stays well below the raw measurements, despite relatively high measurement noise. Fig. 6 shows closer views of the performance measures for three maneuvers: the $-30^{\circ}/s$ sharp turn (see segments D and E), the $80^{\circ}/s$ sharp turn (see segments L and M), and the $5^{\circ}/s$ slow turn (see segment H). IMM-CT-S does have the best performance during nonmaneuvering segments, but IMM-CT-S and IMM-CT-F do not show comparable RMSE reduction after sharp maneuvers.

It can be seen from the results that when it comes to position and velocity error, the IMM-WNTR is a very accurate filter. It is slightly outperformed after the first extreme turn maneuver (see segment B), whose 2-s extreme

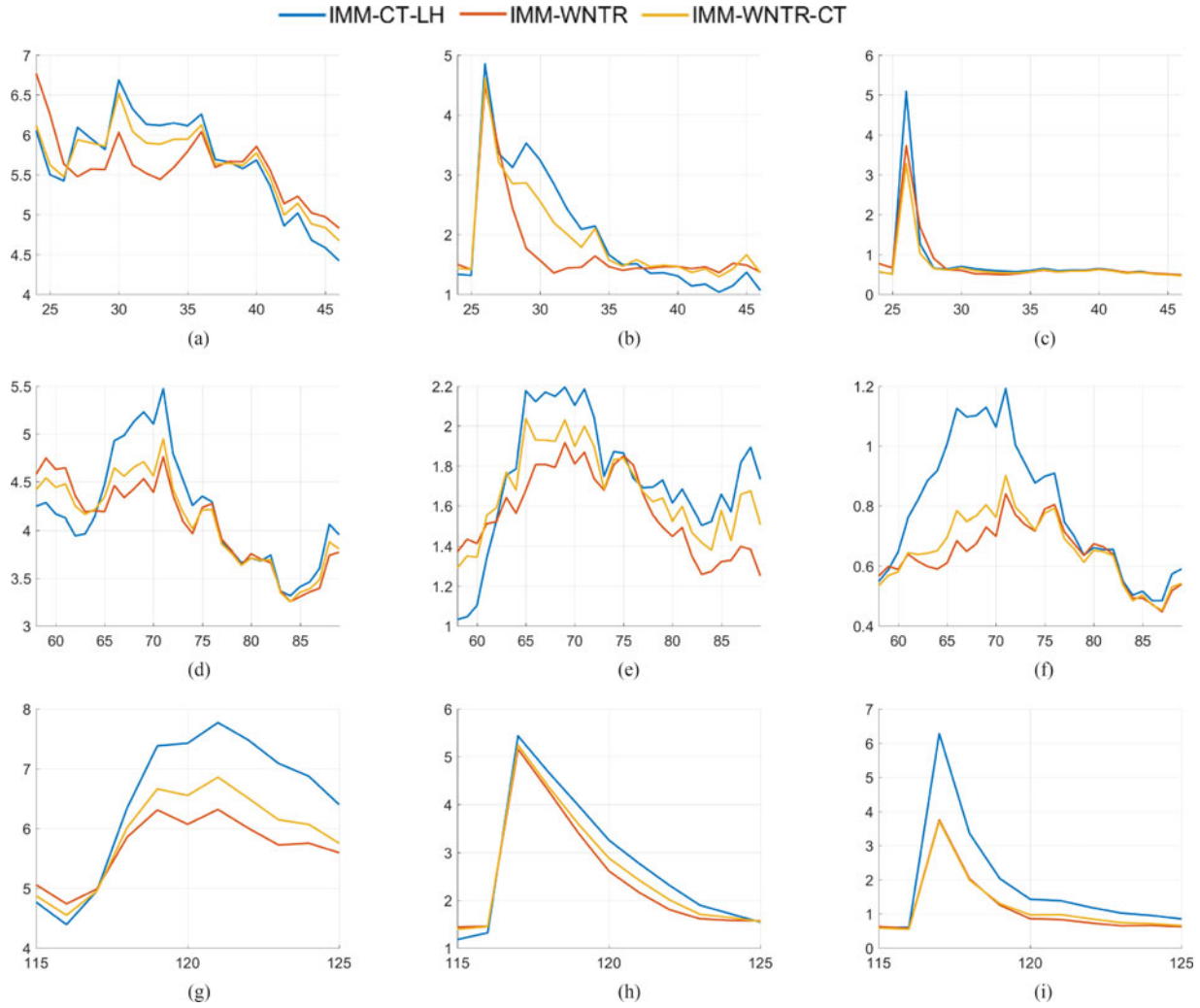


Fig. 6. Magnified plots showing RMSE and NEES during and after three key maneuvers. Axis units are the same as in Figs. 3–5 (a) Pos. RMSE for D, E. (b) Vel. RMSE for D, E. (c) NEES for D, E. (d) Pos. RMSE for H. (e) Vel. RMSE for H. (f) NEES for H. (g) Pos. RMSE for L, M. (h) Vel. RMSE for L, M. (i) NEES for L, M.

turn and simultaneous braking mode is not matched to any of the filters. Apart from IMM-CT-S, which does not perform well after sharp maneuvers, it achieves the lowest RMSE in most other segments (see segments D and E, most of H, and K). It may actually be surprising that it even achieves the best overall performance during the slow CT (see segment H). However, the IMM-WNTR is not a good choice if long-term predictions are needed, as is the case when measurements are slow to arrive. This is because the filter does not attempt to estimate the turn rate ω_k . The NEES of the IMM-WNTR contains sharp peaks after maneuvers but recovers quickly after.

The IMM-CT-LH and the IMM-WNTR-CT are designed to handle both slow and sharp turns, and they both are capable of long-term prediction since they estimate the turn rate. The RMSE of IMM-CT-LH reaches very low values at times (see segments E, G, and I) but it contains higher peaks after maneuvers (see segments D, H, and L).

Similar arguments can be made about its NEES. The IMM-WNTR-CT, designed to have the capability to track during sharp or slow turns, exhibits low RMSE (almost as good as IMM-WNTR), and it provides $\hat{\omega}_{k|k}$. It can be seen that it does not outperform IMM-CT-LH during straight-line motion, but it has better RMSE and NEES after all maneuvers.

C. Turn Rate Estimate

It is worth noting that IMM-WNTR-CT is capable of estimating the turn rate ω_k . In Fig. 7, we see that this is true, although IMM-CT-LH seems to react to the change in ω faster. This assures that IMM-WNTR-CT can be used for long-term prediction if measurements are delayed.

D. Inspection of Mode Probabilities

In Figs. 8–10, we plot the mode probabilities for IMM-CT-S (a conventional IMM-CT), IMM-CT-F (an IMM-CT with zero probability of remaining in the maneuvering

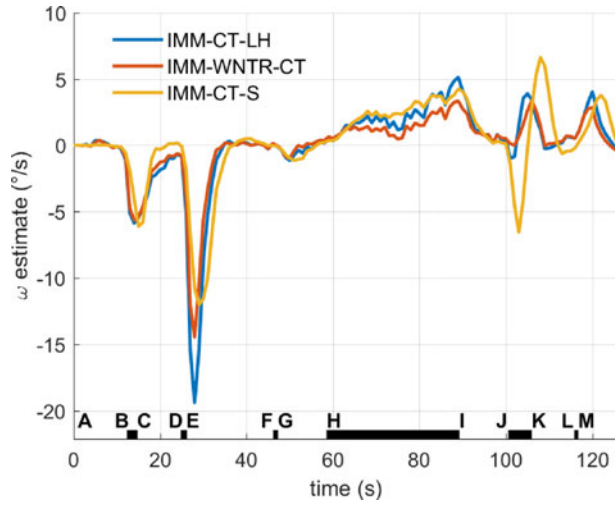


Fig. 7. Average $\hat{\omega}_{k|k}$ from three filters.

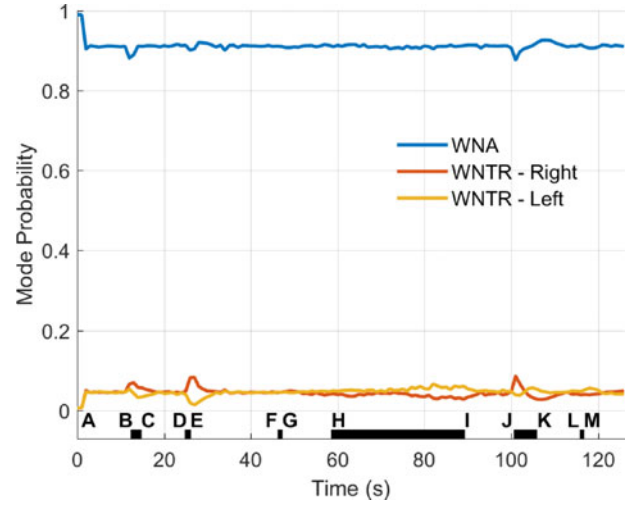


Fig. 10. IMM-WNTR average mode probabilities.

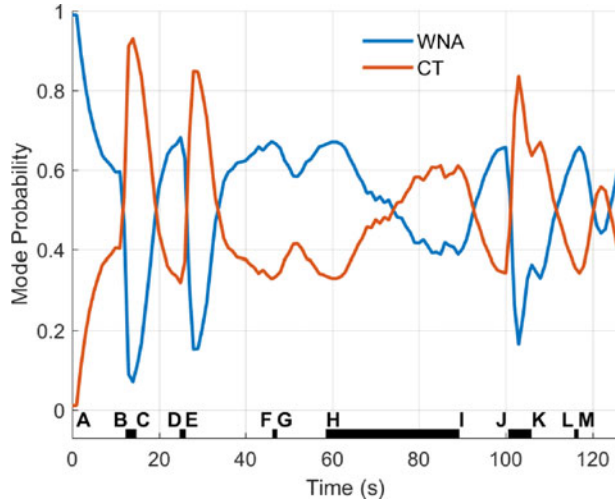


Fig. 8. IMM-CT-S average mode probabilities.

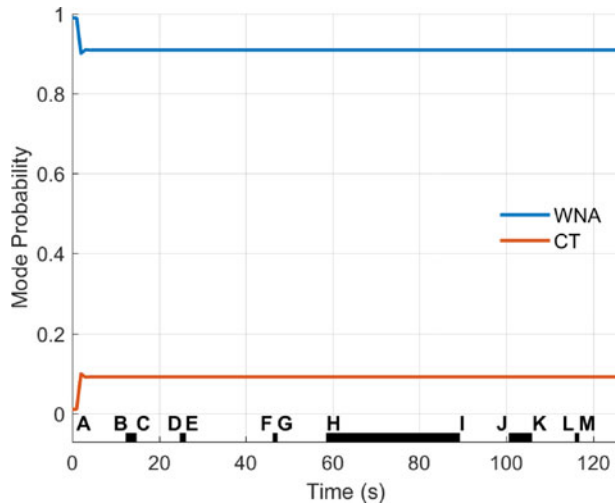


Fig. 9. IMM-CT-F average mode probabilities.

mode), and IMM-WNTR, respectively. We show these as examples of how the IMM treats short-duration maneuvers in terms of its maneuver detection capability. The mode probabilities for IMM-CT-S vary considerably compared to the other two because of the specified long mean sojourn time $E[\tau_2]$, which is required for conventional IMM-CT algorithms. For IMM-CT-F, we see that since $p_{22} = 0$ and because of the random walk turn rate of the discrete CT model, the algorithm is never capable of detecting a maneuver. In the IMM-WNTR, which also has zero probability of returning to left- or right-turn modes, we can still see small perturbations in the mode probabilities (see Fig. 10). These small changes in mode probabilities are enough to open up the filter bandwidth when maneuvers are detected.

E. Computational Complexity

The three-mode IMM-WNTR and the four-mode IMM-WNTR-CT are computationally efficient in the sense that the only extra computation needed, when compared to a single-mode filter or a two-mode IMM, is having to run additional filters through the IMM, and this scales between linearly ($O(n_r)$) and quadratically ($O(n_r^2)$) with the number of additional modes, since each nonlinear filter only runs once per measurement arrival in an IMM, but the mixing step requires interaction between every pair of modes. As $n_r \rightarrow \infty$, ($O(n_r^2)$) will dominate.

As an example, running 1000 Monte Carlo simulations on a single-threaded MATLAB platform using a 32-b, 2.9 GHz system with 3 GB of RAM, the average computation times per measurement interval are: 5.6 ms for IMM-CT-S; 3.9 ms for IMM-CT-F; 10.5 ms for IMM-CT-LH; 18.6 ms for IMM-WNTR; and 24.0 ms for IMM-WNTR-CT.

VI. CONCLUSION

This paper presents a novel nonzero mean, white noise turn rate dynamic model that can be used within a MM estimator to track ground vehicles, moving in an open field,

in two dimensions. Due to their nature, some targets, such as land vehicles with continuous wheel tracks, are capable of performing sharp-turn maneuvers as high as $60^\circ/\text{s}$ while maintaining almost constant speed, and this presents a challenge for conventional state estimator designs, including MM methods that were originally designed for targets with relatively slow CT capability. The nature of the short-lived maneuvers requires the design of transition probability matrices that have a distinct form compared to those for tracking CT targets. Having sharp left- and right-turn modes in an IMM with the TPMs described in this paper is similar to specifying a multimodal turn rate process noise PDF, and the result is a filter that performs extremely well in terms of position and velocity RMSE at the expense of not being able to perform accurate long-term prediction. When also including a slow-turn CT mode in the IMM-WNTR filter, an estimate of the turn rate is provided and long-term predictions can be more accurate. The proposed filters are computationally inexpensive, and any nonlinear filtering algorithm, as well as any MM estimator, can be used with the dynamic models presented here.

APPENDIX

A simple derivation of the Jacobian and Hessian matrices for use in the second-order EKF with both the CT and WNTR models is shown here. We start with the discretized velocity $\dot{\mathbf{s}}$ and position $\mathbf{s}$ evolutions from (7) as

$$\dot{\mathbf{s}}_{k+1} = e^{\mathbf{U}\omega_k T} \dot{\mathbf{s}}_k \quad (17)$$

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \mathbf{U}\omega_k^{-1} (\mathbf{I} - e^{\mathbf{U}\omega_k T}) \dot{\mathbf{s}}_k. \quad (18)$$

The vectors of partial derivatives are

$$\frac{\partial \dot{\mathbf{s}}_{k+1}}{\partial \omega_k} = \mathbf{U}T e^{\mathbf{U}\omega_k T} \dot{\mathbf{s}}_k \quad (19)$$

$$\frac{\partial \mathbf{s}_{k+1}}{\partial \omega_k} = [\omega_k^{-1} T e^{\mathbf{U}\omega_k T} - \mathbf{U}\omega_k^{-2} (\mathbf{I} - e^{\mathbf{U}\omega_k T})] \dot{\mathbf{s}}_k. \quad (20)$$

This leads to the Jacobian of the CT model

$$\frac{\partial \mathbf{f}_{k+1}}{\partial \mathbf{x}_k}_{\text{CT}} = \begin{bmatrix} \mathbf{I}_{2 \times 2} & \frac{1}{\omega} \mathbf{U}(\mathbf{I} - \mathbf{V}_k) & \frac{\partial \mathbf{s}_{k+1}}{\partial \omega_k} \\ \mathbf{0}_{2 \times 2} & \mathbf{V}_k & \frac{\partial \dot{\mathbf{s}}_{k+1}}{\partial \omega_k} \\ \mathbf{0}_{1 \times 2} & \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}_{\omega_k = \hat{\omega}_{k|k}}. \quad (21)$$

The WNTR model is nonlinear in its process noise input only, so

$$\frac{\partial \mathbf{f}_{k+1}}{\partial \mathbf{v}_k}_{\text{WNTR}} = \begin{bmatrix} \frac{T^2}{2} \mathbf{I}_{2 \times 2} & \frac{\partial \mathbf{s}_{k+1}}{\partial \omega_k} \\ T \mathbf{I}_{2 \times 2} & \frac{\partial \dot{\mathbf{s}}_{k+1}}{\partial \omega_k} \end{bmatrix}_{\omega_k = \hat{\omega}_k}. \quad (22)$$

Next, we compute the second-order partial derivatives

$$\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k^2} = -T^2 e^{\mathbf{U}\omega_k T} \dot{\mathbf{s}}_k \quad (23)$$

$$\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} = \mathbf{U}T e^{\mathbf{U}\omega_k T} \quad (24)$$

$$\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k^2} = \left[\frac{2}{\omega_k^3} \mathbf{U} + \left(\frac{1}{\omega_k} \mathbf{U}T^2 - \frac{2}{\omega_k^2} T - \frac{2}{\omega_k^3} \mathbf{U} \right) e^{\mathbf{U}\omega_k T} \right] \dot{\mathbf{s}}_k \quad (25)$$

$$\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} = \omega_k^{-1} T e^{\mathbf{U}\omega_k T} - \mathbf{U}\omega_k^{-2} (\mathbf{I} - e^{\mathbf{U}\omega_k T}). \quad (26)$$

Note that (24) and (26) are 2×2 matrices, whereas (23) and (25) are 2×1 vectors. As explained in [6], an augmented vector for the WNTR model is created as

$$\mathbf{x}_a = [\mathbf{s} \quad \dot{\mathbf{s}} \quad v_\omega]^\top \quad (27)$$

and $\mathbf{x}_a = \mathbf{x}$ in the CT model. The Hessians are

$$\frac{\partial^2 x}{\partial \mathbf{x}_a^2} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \left[\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_1 \\ \mathbf{0}_{1 \times 2} & \left[\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_1' & \left[\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k^2} \right]_1 \end{bmatrix} \quad (28)$$

$$\frac{\partial^2 y}{\partial \mathbf{x}_a^2} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \left[\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_2 \\ \mathbf{0}_{1 \times 2} & \left[\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_2' & \left[\frac{\partial^2 \mathbf{s}_{k+1}}{\partial \omega_k^2} \right]_2 \end{bmatrix} \quad (29)$$

$$\frac{\partial^2 \dot{x}}{\partial \mathbf{x}_a^2} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \left[\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_1 \\ \mathbf{0}_{1 \times 2} & \left[\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_1' & \left[\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k^2} \right]_1 \end{bmatrix} \quad (30)$$

$$\frac{\partial^2 \dot{y}}{\partial \mathbf{x}_a^2} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \left[\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_2 \\ \mathbf{0}_{1 \times 2} & \left[\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k \partial \dot{\mathbf{s}}_k} \right]_2' & \left[\frac{\partial^2 \dot{\mathbf{s}}_{k+1}}{\partial \omega_k^2} \right]_2 \end{bmatrix} \quad (31)$$

where (28)–(31) are evaluated at $\omega_k = \hat{\omega}_{k|k}$ in the CT model and $\omega_k = \bar{v}_\omega$ in the WNTR model, with $[\mathbf{A}]_i$ being the i th row of matrix $\mathbf{A}$ and $[\mathbf{A}]_i'$ the transpose of that row.

ACKNOWLEDGMENT

The authors would like to thank A. Corbeil from TSC for providing the motivation to look into this problem.

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